Status	Finished
Started	Monday, 19 October 2026, 11:15 AM
Completed	Monday, 19 October 2026, 11:19 AM
Duration	3 mins 56 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(log n) ✓
- b. O(n log n)
- c. O👎
- d. O(1)

Question **2**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

- a. Compare the target element with the first element in the list
- b. Sort the list ✗
- c. Compare the target element with the middle element in the list
- d. Divide the list into two equal parts

Question **3**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(log n)
- b. O👎 ✓
- c. O(n log n)
- d. O(1)

Question **4**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering
b. Mining
c. Searching ✓
d. Finding

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search starts over from the beginning
b. The search backtracks to find duplicate elements
c. The search continues until the end of the list
d. The search stops immediately ✓

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the left sublist ✓
b. In the entire list
c. At the beginning of the list
d. In the right sublist

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Quick search
b. Merge search
c. Bubble search
d. Linear search ✓

Question **8**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. n ✓
c. log n
d. n/2

Question **9**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Binary search ✓
b. Hash search
c. Sequential search
d. Linear search

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly
b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
c. Must use a sorted array
d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **11**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all
b. Item is the last element in the array or item is not there at all
c. When the item is somewhere in the middle of the array ✓
d. When the item is the last element in the array

Question **12**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Linear search ✓
- b. Binary search
- c. Jump search
- d. Hashing

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It does not work with negative numbers
- b. It is slower than linear search for small lists
- c. It can only be applied to large lists
- d. It requires the list to be sorted ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Sorting the list before searching
- b. Skipping every second element
- c. Dividing the list in half repeatedly
- d. Checking each element sequentially ✓

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search is more efficient than binary search on large lists.
- b. Linear search requires the list to be sorted.
- c. Linear search divides the list into halves.
- d. Linear search can be applied to both sorted and unsorted lists. ✓